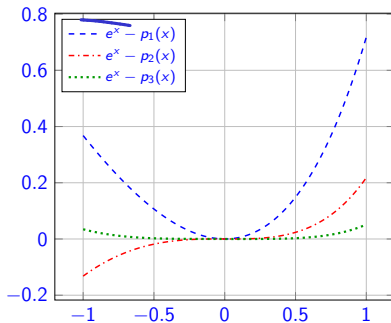
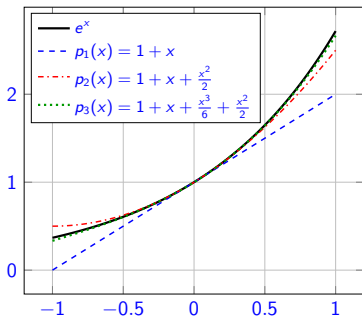


MA385 Part 1: Solving nonlinear equations

1.4: Taylor's Theorem

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0. Outline

1 Taylor's Theorem

■ From MVT to TT

2 The Taylor Polynomial

3 The Remainder

4 An application of Taylor's Theorem

5 Exercises

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Some notation for today:

$f^{(k)}(x)$ is short-hand for $\frac{d^k f}{dx^k}(x)$.

So $f(x) = f^{(0)}(x)$, $f'(x) = f^{(1)}(x)$, $f''(x) = f^{(2)}(x)$, etc.

1. Taylor's Theorem

Taylor's Theorem is perhaps the most important mathematical tool in Numerical Analysis. Providing we can evaluate the derivatives of a given function at some point, it gives us a way of approximating the function by a polynomial.

Working with polynomials, particularly ones of degree 3 or less, is much easier than working with arbitrary functions. For example, polynomials are easy to differentiate and integrate. Most importantly for the next section of this course, their zeros are easy to find.

Brook Taylor, 1685 – 1731, England. He (re)discovered this polynomial approximation in 1712, though its full importance was not realised for another 50 years.



In Section 1.3 we had the **Mean Value Theorem**: If f is function that is continuous and differentiable on the interval $[a, x]$, then there is a point $c \in [a, x]$ such that

$$\frac{f(x) - f(a)}{x - a} = f'(c).$$

As noted, the MVT tells us that we can approximate the value of a function by a near-by value, with accuracy that depends on f' :

$$f(x) = f(a) + f'(c)(x - a).$$


Error :

depends on
 $|f'(c)|$ & $|x - a|$

What if we want a better approximation? We could replace our function, $f(x)$, with a quadratic polynomial. For example, let

$$p_2(x) = b_0 + b_1(x - a) + b_2(x - a)^2,$$

and solve for the coefficients b_0 , b_1 and b_2 so that

$$\underbrace{f(a) = p_2(a)}_{\textcircled{1}}, \quad \underbrace{f'(a) = p_2'(a)}_{\textcircled{2}}, \quad f''(a) = p_2''(a).$$

$$\begin{aligned} \textcircled{1}: p_2(a) &= f(a) \Rightarrow b_0 + b_1(a-a) + b_2(a-a)^2 = f(a) \Rightarrow b_0 = f(a). \\ \textcircled{2}: p_2'(a) &= f'(a) \Rightarrow b_1 + 2b_2(a-a) = f'(a) \Rightarrow b_1 = f'(a). \\ \textcircled{3}: p_2''(a) &= f''(a) \Rightarrow 2b_2 = f''(a) \Rightarrow b_2 = \frac{1}{2}f''(a). \end{aligned}$$

[Finished here Tue 17 Sep (w2.2)]

Next, if we try to construct an approximating cubic of the form

$$\begin{aligned} p_3(x) &= b_0 + b_1(x - a) + b_2(x - a)^2 + b_3(x - a)^3, \\ &= \sum_{k=0}^3 b_k(x - a)^k, \end{aligned}$$

with the property that

$$\begin{aligned} p_3(a) &= f(a), & p'_3(a) &= f'(a), \\ p''_3(a) &= f''(a), & p'''_3(a) &= f'''(a). \end{aligned} \tag{1}$$

Again we find that

$$b_k = \frac{f^{(k)}(a)}{k!} \quad \text{for } k = 0, 1, 2, 3.$$

Note: $f^{(0)}(a) = f(a)$, and $0! = 1$.

2. The Taylor Polynomial

Definition (Taylor Polynomial)

The *Taylor Polynomial* of degree k (also called the *Truncated Taylor Series*) that approximates the function f about the point $x = a$ is

$$p_k(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2}f''(a) + \frac{(x - a)^3}{3!}f'''(a) + \cdots + \frac{(x - a)^k}{k!}f^{(k)}(a).$$

Note: this is a polynomial of degree k .

Also, the $f^{(k)}(a)$ terms are all constants.

2. The Taylor Polynomial

Example

Write down the Taylor polynomial of degree k that approximates $f(x) = e^x$ about the point $x = 0$.

we have that $p_1(x) = f(a) + (x-a)f'(a)$.
with $a=0$, so $p_1(x) = 1 + x$.

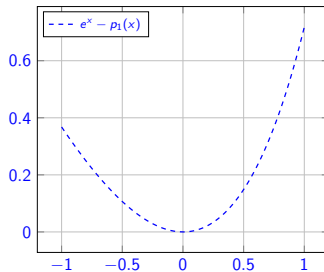
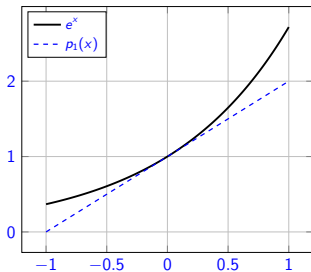


Figure 1: Taylor polys for $f(x) = e^x$ about $x = 0$ (left), and errors (right)

2. The Taylor Polynomial

Example

Write down the Taylor polynomial of degree k that approximates $f(x) = e^x$ about the point $x = 0$.

If $k=2$, we get $p_2(x) = 1 + x + \frac{1}{2}x^2$

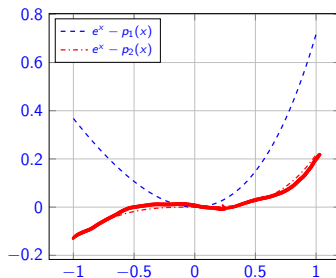
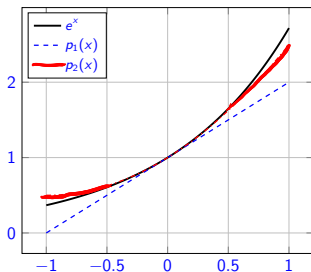


Figure 1: Taylor polys for $f(x) = e^x$ about $x = 0$ (left), and errors (right)

2. The Taylor Polynomial

Example

Write down the Taylor polynomial of degree k that approximates $f(x) = e^x$ about the point $x = 0$.

$$k=3$$

$$p_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$$

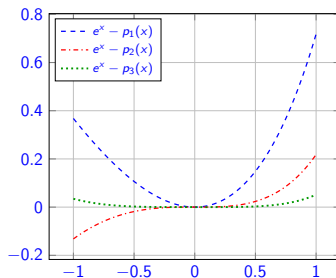
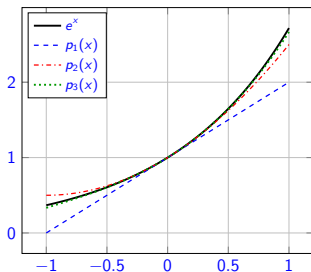


Figure 1: Taylor polys for $f(x) = e^x$ about $x = 0$ (left), and errors (right)

3. The Remainder

We now want to examine the *accuracy* of the Taylor polynomial as an approximation. In particular, we would like to find a formula for the *remainder* or *error*:

$$R_k(x) := f(x) - p_k(x).$$

With a little bit of effort one can prove that:

$$R_k(x) := \frac{(x-a)^{k+1}}{(k+1)!} f^{(k+1)}(\sigma), \text{ for some } \sigma \in [x, a].$$

↖ note: not a!

We won't prove this in class. But for the sake of completeness, I'll add it as an appendix to these notes later.

3. The Remainder

Example

With $f(x) = e^x$ and $a = 0$, we get that

$$R_k(x) = \frac{x^{k+1}}{(k+1)!} e^\sigma, \text{ some } \sigma \in [0, x].$$

Example

How many terms are required in the Taylor Polynomial for e^x about $x = 0$ to ensure that the error at $x = 1$ is

- ▶ no more than 10^{-1} ? $k=5$
- ▶ no more than 10^{-2} ? $k=6$
- ▶ no more than 10^{-6} ? $k=10$
- ▶ no more than 10^{-10} ?

Idea: check
 $|R_k(1)| \leq \frac{1}{k!} e$
for $k=1, 2, 3, 4, \dots$

4. An application of Taylor's Theorem

Taylor's theorem is important to MA385 because...

- ▶ We'll use it to derive, and analyse, Newton's method, which is the most important method for solving nonlinear equations (at least for problems where we can evaluate derivatives).
- ▶ It is the basis for the methods in Section 2 for solving initial value differential equations.

Finished 21 Sep.

5. Exercises

Exercise 1.4.1

Write down the formula for the Taylor Polynomial for

- (i) $f(x) = 3x^2 + 3x - 12$
- (ii) $f(x) = \sqrt{1+x}$ about the point $a = 0$,
- (iii) $f(x) = \log(x)$ about the point $a = 1$.

Exercise 1.4.2

Write out the Taylor polynomial at x , about $a = 0$, of degree 7 for $f(x) = \sin(x)$. How does its derivative compare to the corresponding Taylor polynomial for $f(x) = \cos(x)$?

The purpose of the next exercise is to demonstrate that, usually, the closer x is to a , the better the Taylor polynomial approximates that function's value.

5. Exercises

Exercise 1.4.3

Write out the Taylor Polynomial about $a = 1$ of degree 4 and corresponding remainder for $f(x) = \ln(x)$. Give an upper bound for this remainder when $x = 2$, $x = 1.1$ and $x = 1.01$.